
Physics 523 HW 6

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CONTENTS

I Problem 1	2
1. Critical Point: $\phi = 4$	2
2. $\phi = 0$	2
3. $0 < \phi < 4$	3
4. $4 < \phi$	3
5. Physics	5
II Problem 2	6
6. Lagrangian	6
7. Feynman Rules	6
8. $\chi \rightarrow \chi$ Matrix Element	6
8.1. Feynman Parameters	6
8.2. Complete the Square	7
8.3. Shift Integrand	7
8.4. Divergence	7
8.5. DimReg	7
9. Integrate ♠	8
9.1. Complete the Square	8
9.2. $r = 0$	8
9.3. $0 < r < 4$	8
9.4. $r = 4$	8
9.5. $4 < r$	9
10. $\chi \rightarrow \chi$ Matrix Element: Explicit Form	9
10.1. $0 < s/m^2 = r \leq 4$	9
10.2. $4 \leq s/m^2 = r$	9
11. Decay: $\chi \rightarrow \phi\phi$	9
12. Optical Theorem	11

I use the notation in my Real Scalar Field Notes. I usually use p for the magnitude of the momentum 3-vector, $p = |\mathbf{p}|$, so $p^2 = \mathbf{p} \cdot \mathbf{p}$. But for this homework I will use p^2 for the dot product of the momentum 4-vector: $p^2 = \vec{p} \cdot \vec{p} = \vec{p}^2 = (p^0)^2 - \mathbf{p}^2$.

Part I

Problem 1

$$\spadesuit(s) = \frac{2\alpha}{\pi} \int_0^1 dx x(1-x) \ln \left[\frac{m^2 - x(1-x)s}{m^2} \right] \quad (1)$$

$$\clubsuit(\phi) = \int_0^1 dx x(1-x) \ln[1 - x(1-x)\phi] \quad (2)$$

$$\boxed{\spadesuit(s) = \frac{2\alpha}{\pi} \clubsuit\left(\frac{s}{m^2}\right)} \quad (3)$$

We deal with $\clubsuit(\phi)$ instead of $\spadesuit(s)$. It is simpler and the two are connected by Eq. 3.

$$0 \leq \phi < 4 \implies 0 \leq s < 4m^2, \quad \phi = 4 \implies s = 4m^2, \\ 4 < \phi \implies 4m^2 < s \quad (4)$$

1. CRITICAL POINT: $\phi = 4$

The integrand in Eq. 2 is real if

$$1 - x(1-x)\phi > 0 \implies \frac{1}{\phi} > x(1-x). \quad (5)$$

For $0 < x < 1$ this inequality holds for $0 < \phi < 4$. Below/above the critical point $\clubsuit(\phi)$ is real/complex. It's not obvious if the critical point is real without integrating it (which I now do).

$$\clubsuit(4) = \int_0^1 dx x(1-x) \ln[1 - 4x(1-x)] \quad (6)$$

$$\clubsuit(4) = \int_0^1 dx x(1-x) \ln(1 - 4x + 4x^2) \quad (7)$$

$$1 - 4x + 4x^2 = (1 - 2x)^2 \quad (8)$$

$$\clubsuit(4) = \int_0^1 dx x(1-x) \ln[(1 - 2x)^2] \quad (9)$$

$$\clubsuit(4) = \int_0^1 dx x(1-x) \ln|1 - 2x|^2 \quad (10)$$

$$\clubsuit(4) = 2 \int_0^1 dx x(1-x) \ln|1 - 2x| \quad (11)$$

$$\frac{\clubsuit(4)}{2} = \int_0^{1/2} dx x(1-x) \ln|1 - 2x| \\ + \int_{1/2}^1 dx x(1-x) \ln|1 - 2x| \quad (12)$$

$$\frac{\clubsuit(4)}{2} = \int_0^{1/2} dx x(1-x) \ln(1 - 2x) \\ + \int_{1/2}^1 dx x(1-x) \ln(2x - 1) \quad (13)$$

$$\frac{\clubsuit(4)}{2} = \star[1] + \star[2] \quad (14)$$

$$\star[1] = \int_0^{1/2} dx x(1-x) \ln(1 - 2x), \quad (15)$$

$$\star[2] = \int_{1/2}^1 dx x(1-x) \ln(2x - 1)$$

$$\star[1] = \int_0^{1/2} dx x(1-x) \ln(1 - 2x), \quad (16)$$

$$y = 1 - 2x, \quad y(0) = 1, \quad y\left(\frac{1}{2}\right) = 0, \quad dx = -\frac{dy}{2}, \quad x = \frac{1-y}{2} \quad (17)$$

$$\star[1] = \frac{1}{8} \int_0^1 dy (1-y)(1+y) \ln y \quad (18)$$

$$\star[1] = \frac{\star}{8}, \quad \star = \int_0^1 dy (1-y^2) \ln y \quad (19)$$

$$\star[2] = \int_{1/2}^1 dx x(1-x) \ln(2x - 1) \quad (20)$$

$$y = 2x - 1, \quad y\left(\frac{1}{2}\right) = 0, \quad y(1) = 1, \quad dx = \frac{dy}{2}, \quad x = \frac{y+1}{2} \quad (21)$$

$$\star[2] = \frac{1}{8} \int_0^1 dy (1-y^2) \ln y = \frac{\star}{8} = \star[1] \quad (22)$$

$$2\clubsuit(4) = \star \quad (23)$$

I integrate $\star$ by hand (it's fun):

$$\star = -\frac{8}{9}. \quad (24)$$

$$\boxed{\clubsuit(4) = -\frac{4}{9}} \quad (25)$$

2. $\phi = 0$

$$\clubsuit(0) = \int_0^1 dx x(1-x) \ln 1, \quad \ln 1 = 0 \quad (26)$$

$$\boxed{\clubsuit(0) = 0} \quad (27)$$

3. $0 < \phi < 4$

$$4\clubsuit(\phi) = \frac{2}{3} \ln \left(\frac{\phi}{4} \right) + \heartsuit \quad (42)$$

$$\clubsuit(\phi) = \int_0^1 dx x(1-x) \ln[1-x(1-x)\phi] \quad (28)$$

$$\heartsuit = \int_0^1 dx \ln(x^2+a) - \int_0^1 dx x^2 \ln(x^2+a) \quad (43)$$

$$\clubsuit(\phi) = \int_0^1 dx x(1-x) \ln(1-\phi x + \phi x^2) \quad (29)$$

$$\int_0^1 dx \ln(x^2+a) = \ln(1+a) + 2\sqrt{a} \operatorname{atan} \left(\frac{1}{\sqrt{a}} \right) - 2, \quad a > 0 \quad (44)$$

I complete the square.

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a} \quad (30)$$

$$9 \int_0^1 dx x^2 \ln(x^2+a) = 6a - 6a^{3/2} \operatorname{acot} \sqrt{a} + 3 \ln(1+a) - 2, \quad a > 0 \quad (45)$$

$$1 - \phi x + \phi x^2 = \phi \left(x - \frac{1}{2} \right)^2 - \frac{\phi}{4} + 1 \quad (31)$$

$$\frac{3}{2} \heartsuit = \ln(1+a) + (3+a) \sqrt{a} \operatorname{acot} \sqrt{a} - a - \frac{8}{3} \quad (46)$$

$$\clubsuit(\phi) = \int_0^1 dx x(1-x) \ln \left[\phi \left(x - \frac{1}{2} \right)^2 - \frac{\phi}{4} + 1 \right] \quad (32)$$

$$\frac{3}{2} \heartsuit = \ln \left(\frac{4}{\phi} \right) + 2 \left(1 + \frac{2}{\phi} \right) \sqrt{\frac{4}{\phi} - 1} \operatorname{acsc} \left(\frac{2}{\sqrt{\phi}} \right) - \frac{4}{\phi} - \frac{5}{3} \quad (47)$$

$$y = x - \frac{1}{2}, \quad y(0) = -\frac{1}{2}, \quad y(1) = \frac{1}{2}, \quad dy = dx, \quad x = y + \frac{1}{2} \quad (33)$$

$$3\clubsuit(\phi) = \left(1 + \frac{2}{\phi} \right) \sqrt{\frac{4}{\phi} - 1} \operatorname{acsc} \left(\frac{2}{\sqrt{\phi}} \right) - \frac{2}{\phi} - \frac{5}{6} \quad (48)$$

$$\clubsuit(\phi) = \frac{1}{4} \int_{-1/2}^{1/2} dy (1-4y^2) \ln \left(\phi y^2 - \frac{\phi}{4} + 1 \right) \quad (34)$$

4. $4 < \phi$

$$\clubsuit(\phi) = \frac{1}{4} \int_{-1/2}^{1/2} dy (1-4y^2) \ln \left[\frac{\phi}{4} \left(4y^2 - 1 + \frac{4}{\phi} \right) \right] \quad (35)$$

$$\clubsuit(\phi) = \int_0^1 dx x(1-x) \ln \left[\phi \left(\frac{1}{\phi} - x + x^2 \right) \right] \quad (49)$$

$$x = 2y, \quad x \left(\pm \frac{1}{2} \right) = \pm 1, \quad \frac{dx}{2} = dy \quad (36)$$

$$\begin{aligned} \clubsuit(\phi) &= \ln \phi \int_0^1 dx x(1-x) \\ &+ \int_0^1 dx x(1-x) \ln \left(\frac{1}{\phi} - x + x^2 \right) \end{aligned} \quad (50)$$

$$\clubsuit(\phi) = \frac{1}{8} \int_{-1}^1 dx (1-x^2) \ln \left[\frac{\phi}{4} \left(x^2 - 1 + \frac{4}{\phi} \right) \right] \quad (37)$$

$$\int_0^1 dx x(1-x) = \frac{1}{6} \quad (51)$$

$$\clubsuit(\phi) = \frac{1}{4} \int_0^1 dx (1-x^2) \ln \left[\frac{\phi}{4} \left(x^2 - 1 + \frac{4}{\phi} \right) \right] \quad (38)$$

$$\diamond = \int_0^1 dx x(1-x) \ln \left(\frac{1}{\phi} - x + x^2 \right) \quad (52)$$

$$4\clubsuit(\phi) = \ln \left(\frac{\phi}{4} \right) \int_0^1 dx (1-x^2) + \heartsuit \quad (39)$$

$$\clubsuit(\phi) = \frac{\ln \phi}{6} + \diamond \quad (53)$$

$$\heartsuit = \int_0^1 dx (1-x^2) \ln(x^2+a), \quad a = \frac{4}{\phi} - 1 > 0 \quad (40)$$

$$\int_0^1 dx (1-x^2) = \frac{2}{3} \quad (41)$$

$$\frac{1}{\phi} - x_{\pm} + x_{\pm}^2 = 0 \implies x_{\pm} = \frac{1 \pm \Delta}{2}, \quad \Delta = \sqrt{1 - \frac{4}{\phi}} > 0 \quad (54)$$

$$\frac{1}{\phi} - x + x^2 = (x - x_+)(x - x_-) \quad (55)$$

$$\diamond = \int_0^1 dx x(1-x) \ln[(x - x_+)(x - x_-)] \quad (56)$$

$$4 < \phi \implies 0 < x_- < x_+ < 1 \quad (57)$$

$$\diamond = \mathfrak{C}[1] + \mathfrak{C}[2] + \mathfrak{C}[3] \quad (58)$$

$$\begin{aligned} \mathfrak{C}[1] &= \int_0^{x_-} dx x(1-x) \ln[(x - x_+)(x - x_-)], \\ \mathfrak{C}[2] &= \int_{x_-}^{x_+} dx x(1-x) \ln[(x - x_+)(x - x_-)], \\ \mathfrak{C}[3] &= \int_{x_+}^1 dx x(1-x) \ln[(x - x_+)(x - x_-)] \end{aligned} \quad (59)$$

$$\clubsuit(\phi) = \frac{\ln \phi}{6} + \mathfrak{C}[1] + \mathfrak{C}[2] + \mathfrak{C}[3] \quad (60)$$

For $0 < x < x_-$

$$x - x_+ < 0, \quad x - x_- < 0 \quad (x - x_+)(x - x_-) > 0. \quad (61)$$

$$\mathfrak{C}[1] = \int_0^{x_-} dx x(1-x) \ln[(x - x_+)(x - x_-)] \quad (62)$$

$$\mathfrak{C}[1] = \int_0^{x_-} dx x(1-x) \ln[(x_+ - x)(x_- - x)] \quad (63)$$

$$\begin{aligned} \mathfrak{C}[1] &= \int_0^{x_-} dx x(1-x) \ln(x_+ - x) \\ &\quad + \int_0^{x_-} dx x(1-x) \ln(x_- - x) \end{aligned} \quad (64)$$

$$\begin{aligned} y &= x_+ - x, & x &= x_+ - y, & dx &= -dy, \\ y(0) &= x_+, & y(x_-) &= x_+ - x_- = \Delta \end{aligned} \quad (65)$$

$$\begin{aligned} z &= x_- - x, & x &= x_- - z, & dx &= -dz, \\ z(0) &= x_-, & z(x_-) &= 0 \end{aligned} \quad (66)$$

$$\mathfrak{C}[1] = \mathfrak{Y}[1] + \mathfrak{Y}[2] \quad (67)$$

$$\begin{aligned} \mathfrak{Y}[1] &= \int_{\Delta}^{x_+} dy (x_+ - y)(1 - x_+ + y) \ln y, \\ \mathfrak{Y}[2] &= \int_0^{x_-} dz (x_- - z)(1 - x_- + z) \ln z \end{aligned} \quad (68)$$

$$\begin{aligned} \mathfrak{Y}[1] &= \int_{\Delta}^{x_+} dy (x_+ - y)(x_- + y) \ln y, \\ \mathfrak{Y}[2] &= \int_0^{x_-} dz (x_- - z)(x_+ + z) \ln z \end{aligned} \quad (69)$$

For $x_+ < x < 1$

$$0 < x - x_+ < x - x_- \implies 0 < (x - x_+)(x - x_-). \quad (70)$$

$$\mathfrak{C}[3] = \int_{x_+}^1 dx x(1-x) \ln(x - x_+) + \int_{x_+}^1 dx x(1-x) \ln(x - x_-) \quad (71)$$

$$\begin{aligned} z &= x - x_+, & x &= z + x_+, & dx &= dz, \\ z(x_+) &= 0, & z(1) &= 1 - x_+ = x_- \end{aligned} \quad (72)$$

$$\begin{aligned} y &= x - x_-, & x &= y + x_-, & dx &= dy, \\ y(x_+) &= x_+ - x_- = \Delta, & y(1) &= 1 - x_- = x_+ \end{aligned} \quad (73)$$

$$\begin{aligned} \mathfrak{C}[3] &= \int_0^{x_-} dz (z + x_+)(x_- - z) \ln z \\ &\quad + \int_{\Delta}^{x_+} dy (y + x_-)(x_+ - y) \ln y \end{aligned} \quad (74)$$

$$\mathfrak{C}[3] = \mathfrak{Y}[2] + \mathfrak{Y}[1] \quad (75)$$

For $x_- < x < x_+$

$$x - x_+ < 0 < x - x_- \implies (x - x_+)(x - x_-) < 0. \quad (76)$$

$$\mathfrak{C}[2] = \int_{x_-}^{x_+} dx x(1-x) \ln[(x - x_+)(x - x_-)] \quad (77)$$

$$\begin{aligned} \mathfrak{C}[2] &= \int_{x_-}^{x_+} dx x(1-x) \ln(x - x_+) \\ &\quad + \int_{x_-}^{x_+} dx x(1-x) \ln(x - x_-) \end{aligned} \quad (78)$$

$$\begin{aligned} y &= x - x_-, & x &= y + x_-, & dy &= dx, \\ y(x_-) &= 0, & y(x_+) &= x_+ - x_- = \Delta \end{aligned} \quad (79)$$

$$\mathfrak{C}[2] = \int_{x_-}^{x_+} dx x(1-x) \ln(x - x_+) + \mathfrak{Y}[3] \quad (80)$$

$$\mathfrak{Y}[3] = \int_0^{\Delta} dy (y + x_-)(x_+ - y) \ln y \quad (81)$$

$$\mathfrak{C}[2] = \int_{x_-}^{x_+} dx x(1-x) \ln(-|x - x_+|) + \mathfrak{Y}[3] \quad (82)$$

5. PHYSICS

$$\begin{aligned} \mathfrak{C}[2] &= \ln(-1) \int_{x_-}^{x_+} dx x(1-x) + \mathfrak{Y}[3] \\ &\quad + \int_{x_-}^{x_+} dx x(1-x) \ln|x - x_+| \quad (83) \end{aligned}$$

$$\int_{x_-}^{x_+} dx x(1-x) = \frac{1}{3} \left(\frac{1}{2} + \frac{1}{\phi} \right) \sqrt{1 - \frac{4}{\phi}} \quad (84)$$

$$\begin{aligned} \mathfrak{C}[2] &= \frac{\ln(-1)}{3} \left(\frac{1}{2} + \frac{1}{\phi} \right) \sqrt{1 - \frac{4}{\phi}} + \mathfrak{Y}[3] \\ &\quad + \int_{x_-}^{x_+} dx x(1-x) \ln(x_+ - x) \quad (85) \end{aligned}$$

$$\begin{aligned} y &= x_+ - x, & x &= x_+ - y, & dx &= -dy, \\ y(x_-) &= x_+ - x_- = \Delta, & y(x_+) &= 0 \end{aligned} \quad (86)$$

$$\begin{aligned} \mathfrak{C}[2] &= \frac{\ln(-1)}{3} \left(\frac{1}{2} + \frac{1}{\phi} \right) \sqrt{1 - \frac{4}{\phi}} + \mathfrak{Y}[3] \\ &\quad + \int_0^\Delta dy (x_+ - y)(x_- + y) \ln y \quad (87) \end{aligned}$$

$$\mathfrak{C}[2] = \frac{\ln(-1)}{3} \left(\frac{1}{2} + \frac{1}{\phi} \right) \sqrt{1 - \frac{4}{\phi}} + 2\mathfrak{Y}[3] \quad (88)$$

$$\ln(-1) = \ln e^{\pm i\pi} = \pm i\pi \quad (89)$$

The sign depends on choosing the right branch: I don't know how to do that. Fix the sign by choosing whichever option makes the optical theorem true.

$$\mathfrak{C}[2] = 2\mathfrak{Y}[3] \pm i\frac{\pi}{3} \left(\frac{1}{2} + \frac{1}{\phi} \right) f(\phi), \quad f(\phi) = \sqrt{1 - \frac{4}{\phi}} \quad (90)$$

$$\clubsuit(\phi) = \frac{\ln \phi}{6} + 2(\mathfrak{Y}[1] + \mathfrak{Y}[2] + \mathfrak{Y}[3]) \pm i\frac{\pi}{3} \left(\frac{1}{2} + \frac{1}{\phi} \right) f(\phi) \quad (91)$$

The $\mathfrak{Y}$ integrals are real and easy to evaluate. But the algebra is tedious so I export it to Mathematica.

$$\clubsuit(\phi) = \text{Re } \clubsuit(\phi) + i \text{Im } \clubsuit(\phi) \quad (92)$$

$$\boxed{\text{Im } \clubsuit(\phi) = \pm \frac{\pi}{3} \left(\frac{1}{2} + \frac{1}{\phi} \right) f(\phi)} \quad (93)$$

$$\text{Re } \clubsuit(\phi) = \frac{\ln \phi}{6} + 2(\mathfrak{Y}[1] + \mathfrak{Y}[2] + \mathfrak{Y}[3]) \quad (94)$$

$$\boxed{\text{Re } \clubsuit(\phi) = \frac{\phi + 2}{6\phi} f(\phi) \ln \left[\frac{1 + f(\phi)}{1 - f(\phi)} \right] - \frac{12 + 5\phi}{18\phi}} \quad (95)$$

The integral in Eq. 1 comes from the leading loop correction to the photon self-energy. The optical theorem connects a nonzero imaginary part to a possible photon decay. We have shown that $\text{Im } \spadesuit(s) = 0$ for $0 \leq s \leq 4m^2$. This makes sense. Remember that the Mandelstam variable $s = p^2$ where $\vec{p}$ is the photon momentum. The energy of a stationary electron-positron pair is $2m$: $s = p^2 = (2m)^2 = 4m^2$. A real (on-shell) photon cannot decay to a single pair because that violates momentum conservation (easiest to see in the center of mass frame). A virtual (off-shell) photon can decay into a single pair (as long as the energy is even slightly above $2m$), which is why $\text{Im } \spadesuit(s) > 0$ for $s > 4m^2$. The critical point behavior implies that even a virtual photon cannot decay into a single pair with energy $2m$, but that's not obvious without doing the calculation. Neither real nor virtual photons can decay into a single electron (positron) because that violates charge conservation.

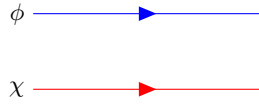
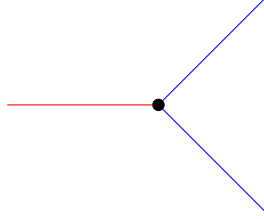


FIG. 1. Propagators


 FIG. 2. Vertex coefficient is $-2ig$.

Part II

Problem 2

6. LAGRANGIAN

I will work with the bare Lagrangian: I won't renormalize.

$$\mathcal{L} = \frac{(\partial\phi)^2 - m^2\phi^2 + (\partial\chi)^2 - \mu^2\chi^2}{2} - g\phi^2\chi \quad (96)$$

$$(\partial\phi)^2 = \eta^{\alpha\beta}(\partial_\alpha\phi)(\partial_\beta\phi) \quad (97)$$

The free Lagrangian is

$$2\mathcal{L}_0 = (\partial\phi)^2 - m^2\phi^2 + (\partial\chi)^2 - \mu^2\chi^2. \quad (98)$$

$$\mathcal{L} = \mathcal{L}_0 - g\phi^2\chi \quad (99)$$

7. FEYNMAN RULES

The propagators (Fig. 1) are

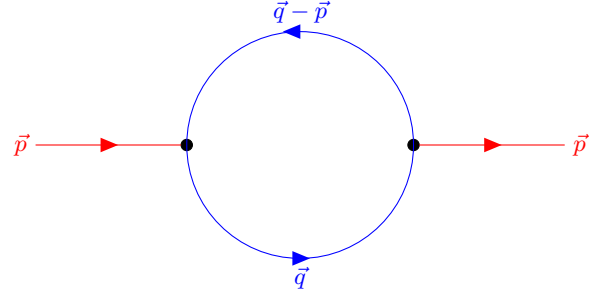
$$\phi \sim \frac{i}{p^2 - m^2 + i\epsilon}, \quad \chi \sim \frac{i}{p^2 - \mu^2 + i\epsilon}. \quad (100)$$

The interaction vertex is in Fig. 2.

8. $\chi \rightarrow \chi$ MATRIX ELEMENT

The trivial part of the S-matrix has 1 diagram (Fig. 3). The non-trivial part, $\mathcal{M}$, has 1 diagram to leading order (Fig. 4).

$$i\mathcal{M} = i\Pi \quad (101)$$


 FIG. 3. Trivial scattering: χ particle propagates freely.

 FIG. 4. χ self-energy loop: $i\Pi$

$$i\mathcal{M} = \frac{S}{(2\pi)^4} (-2ig)^2 \int d^4q \frac{i}{q^2 - m^2 + i\epsilon} \frac{i}{(\vec{q} - \vec{p})^2 - m^2 + i\epsilon} \quad (102)$$

The symmetry factor S (Fig. 5) is

$$S = \frac{2^2}{(2!)^2 2!} = \frac{1}{2}. \quad (103)$$

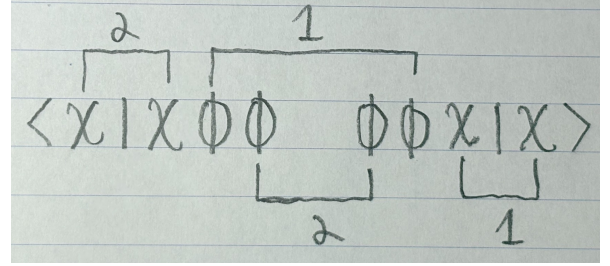


FIG. 5. Symmetry factor combinatorics for Fig. 4.

$$\mathcal{M} = -\frac{2ig^2}{(2\pi)^4} \int \frac{d^4q}{(q^2 - m^2 + i\epsilon)[(\vec{q} - \vec{p})^2 - m^2 + i\epsilon]} \quad (104)$$

8.1. Feynman Parameters

$$A = (\vec{q} - \vec{p})^2 - m^2 + i\epsilon, \quad B = q^2 - m^2 + i\epsilon \quad (105)$$

$$\mathcal{M} = -\frac{2ig^2}{(2\pi)^4} \int \frac{d^4q}{AB} \quad (106)$$

$$\frac{1}{AB} = \int_0^1 \frac{dx}{D^2}, \quad D = A + (B - A)x \quad (107)$$

$$\mathcal{M} = \frac{2g^2}{i(2\pi)^4} \int d^4q \int_0^1 \frac{dx}{D^2} = \frac{2g^2}{i(2\pi)^4} \int_0^1 dx \int \frac{d^4q}{D^2} \quad (108)$$

$$D = (\vec{q} - \vec{p})^2 + [q^2 - (\vec{q} - \vec{p})^2]x - m^2 + i\epsilon \quad (109)$$

$$D = q^2 + p^2 - 2\vec{q} \cdot \vec{p} + (2\vec{q} \cdot \vec{p} - p^2)x - m^2 + i\epsilon \quad (110)$$

$$D = q^2 - 2(1-x)\vec{q} \cdot \vec{p} + (1-x)p^2 - m^2 + i\epsilon \quad (111)$$

8.2. Complete the Square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a} \quad (112)$$

$$x \rightarrow \vec{q}, \quad a \rightarrow 1, \quad b \rightarrow -2(1-x)\vec{p}, \quad c \rightarrow (1-x)p^2 - m^2 + i\epsilon \quad (113)$$

$$D = [\vec{q} - (1-x)\vec{p}]^2 - \Delta + i\epsilon \quad (114)$$

$$\Delta = m^2 - (1-x)xp^2 \quad (115)$$

$$\mathcal{M} = \frac{2g^2}{i(2\pi)^4} \int_0^1 dx \int \frac{d^4q}{([\vec{q} - (1-x)\vec{p}]^2 - \Delta + i\epsilon)^2} \quad (116)$$

$$\mathcal{M} = \frac{2g^2}{i(2\pi)^4} \int_0^1 dx \int \frac{d^4k}{([\vec{k} - (1-x)\vec{p}]^2 - \Delta + i\epsilon)^2} \quad (117)$$

8.3. Shift Integrand

$$\vec{q} = \vec{k} - (1-x)\vec{p}, \quad d^4k = d^4q \quad (118)$$

$$\mathcal{M} = \frac{2g^2}{i(2\pi)^4} \int_0^1 dx \int \frac{d^4q}{(q^2 - \Delta + i\epsilon)^2} \quad (119)$$

8.4. Divergence

$$\mathcal{M} \sim \int_1^\Lambda dq \frac{q^3}{q^4} = \int_1^\Lambda \frac{dq}{q} = \ln \Lambda, \quad \Lambda \rightarrow \infty \quad (120)$$

8.5. DimReg

$$I = \frac{1}{(2\pi)^4} \int \frac{d^4q}{(q^2 - \Delta + i\epsilon)^2}, \quad \mathcal{M} = -2ig^2 \int_0^1 dx I \quad (121)$$

$$I(d) = \frac{1}{(2\pi)^d} \int \frac{d^d q}{(q^2 - \Delta + i\epsilon)^2} \quad (122)$$

In my Real Scalar Field Notes I derive

$$\int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^2} = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right). \quad (123)$$

$$I(d) = \frac{i\Gamma(2-d/2)}{2^d \pi^{d/2} \Delta^{2-d/2}} = \frac{i}{\Delta^2} \left(\frac{\sqrt{\Delta/\pi}}{2} \right)^d \Gamma\left(2 - \frac{d}{2}\right) \quad (124)$$

$$I = I(4) = I(4 - \epsilon), \quad \epsilon \rightarrow 0 \quad (125)$$

$$\mathcal{M} = -2ig^2 \int_0^1 dx I(4 - \epsilon) \quad (126)$$

$$\mathcal{M} = 2g^2 \int_0^1 dx \frac{1}{\Delta^2} \left(\frac{\sqrt{\Delta/\pi}}{2} \right)^{4-\epsilon} \Gamma\left(\frac{\epsilon}{2}\right) \quad (127)$$

$$\mathcal{M} = \frac{g^2}{8\pi^2} \int_0^1 dx \left(\frac{2}{\sqrt{\Delta/\pi}} \right)^\epsilon \Gamma\left(\frac{\epsilon}{2}\right) \quad (128)$$

This expression yields a dimensionful log. I don't like that. I introduce a scale $[\beta] = 1$. Because $\beta^\epsilon \rightarrow 1$ as $\epsilon \rightarrow 0$ I can add as many β^ϵ factors as I need.

$$\mathcal{M} = \frac{g^2}{8\pi^2} \int_0^1 dx \left(\frac{2\beta}{\sqrt{\Delta/\pi}} \right)^\epsilon \Gamma\left(\frac{\epsilon}{2}\right) \quad (129)$$

I take $\epsilon \rightarrow 0$.

$$\mathcal{M} = \frac{g^2}{8\pi^2} \int_0^1 dx \left[\frac{2}{\epsilon} - \gamma + \ln\left(\frac{4\pi\beta^2}{\Delta}\right) \right] \quad (130)$$

Here γ is the Euler–Mascheroni constant $\gamma \approx 0.577216$.

$$\mathcal{M} = \frac{g^2}{8\pi^2} \left(\frac{2}{\epsilon} - \gamma + \mathbb{I} \right), \quad \mathbb{I} = \int_0^1 dx \ln\left(\frac{4\pi\beta^2}{\Delta}\right) \quad (131)$$

$$\mathbb{I} = \int_0^1 dx \ln\left(\frac{4\pi\beta^2}{\Delta}\right) \quad (132)$$

$$\mathbb{I} = \int_0^1 dx \ln\left[\frac{4\pi\beta^2}{m^2 - (1-x)xs}\right], \quad s = p^2 \quad (133)$$

$$\mathbb{I} = - \int_0^1 dx \ln \left[\frac{m^2 - (1-x)xs}{4\pi\beta^2} \right] \quad (134)$$

$$\mathbb{I} = - \int_0^1 dx \ln \left[\frac{m^2}{4\pi\beta^2} \left(1 - \frac{(1-x)xs}{m^2} \right) \right] \quad (135)$$

$$\mathbb{I} = \ln \left(4\pi \frac{\beta^2}{m^2} \right) - \int_0^1 dx \ln \left[1 - \frac{(1-x)xs}{m^2} \right] \quad (136)$$

$$\spadesuit(r) = \int_0^1 dx \ln[1 - (1-x)xr] \quad (137)$$

$$\mathbb{I} = \ln \left(4\pi \frac{\beta^2}{m^2} \right) - \spadesuit \left(\frac{s}{m^2} \right) \quad (138)$$

$$\mathcal{M} = \frac{g^2}{8\pi^2} \left[\frac{2}{\epsilon} - \gamma + \ln \left(4\pi \frac{\beta^2}{m^2} \right) - \spadesuit \left(\frac{s}{m^2} \right) \right] \quad (139)$$

9. INTEGRATE ♠

$$\spadesuit(r) = \int_0^1 dx \ln[1 - (1-x)xr] \quad (140)$$

$$\spadesuit(r) = \int_0^1 dx \ln(rx^2 - xr + 1) \quad (141)$$

9.1. Complete the Square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a} \quad (142)$$

$$rx^2 - xr + 1 = r \left(x - \frac{1}{2} \right)^2 - \frac{r}{4} + 1 \quad (143)$$

$$\spadesuit(r) = \int_0^1 dx \ln \left[r \left(x - \frac{1}{2} \right)^2 - \frac{r}{4} + 1 \right] \quad (144)$$

$$y = x - \frac{1}{2}, \quad y(0) = -\frac{1}{2}, \quad y(1) = \frac{1}{2}, \quad dy = dx \quad (145)$$

$$\spadesuit(r) = \int_{-1/2}^{1/2} dy \ln \left(ry^2 - \frac{r}{4} + 1 \right) \quad (146)$$

$$\spadesuit(r) = 2 \int_0^{1/2} dy \ln \left(ry^2 - \frac{r}{4} + 1 \right) \quad (147)$$

$$\spadesuit(r) = 2 \int_0^{1/2} dy \ln \left[\frac{r}{4} \left(4y^2 - 1 + \frac{4}{r} \right) \right] \quad (148)$$

$$x = 2y, \quad \frac{dx}{2} = dy, \quad x(0) = 0, \quad x \left(\frac{1}{2} \right) = 1 \quad (149)$$

$$\spadesuit(r) = \int_0^1 dx \ln \left[\frac{r}{4} \left(x^2 - 1 + \frac{4}{r} \right) \right] \quad (150)$$

$$\spadesuit(r) = \ln \left(\frac{r}{4} \right) + \heartsuit(r), \quad \heartsuit(r) = \int_0^1 dx \ln \left(x^2 - 1 + \frac{4}{r} \right) \quad (151)$$

9.2. $r = 0$

$$\spadesuit(0) = \ln 1 = 0, \quad \boxed{\spadesuit(0) = 0} \quad (152)$$

9.3. $0 < r < 4$

$$\heartsuit(r) = \int_0^1 dx \ln \left(x^2 - 1 + \frac{4}{r} \right) \quad (153)$$

$$\heartsuit(r) = 2\sqrt{\frac{4}{r} - 1} \arcsin \left(\frac{\sqrt{r}}{2} \right) + \ln \left(\frac{4}{r} \right) - 2 \quad (154)$$

$$\spadesuit(r) = 2\sqrt{\frac{4}{r} - 1} \arcsin \left(\frac{\sqrt{r}}{2} \right) - 2 \quad (155)$$

9.4. $r = 4$

$$\heartsuit(4) = 2 \int_0^1 dx \ln x \quad (156)$$

$$\begin{aligned} \int_0^1 dx \ln x &= \int_0^1 dx \frac{dx}{dx} \ln x = - \int_0^1 dx x \frac{d}{dx} \ln x = \\ &= - \int_0^1 dx \frac{x}{x} = - \int_0^1 dx = -1 \end{aligned} \quad (157)$$

$$\boxed{\heartsuit(4) = \spadesuit(4) = -2} \quad (158)$$

9.5. $4 < r$

$$\heartsuit(r) = \int_0^1 dx \ln \left(x^2 - 1 + \frac{4}{r} \right) \quad (159)$$

$$\heartsuit(r) = \int_0^1 dx \ln[(x-a)(x+a)], \quad a = \sqrt{1 - \frac{4}{r}} \quad (160)$$

$$\heartsuit(r) = \int_0^1 dx \ln(x-a) + \int_0^1 dx \ln(x+a) \quad (161)$$

$$\int_0^1 dx \ln(x+a) = \ln(1+a) + a \ln \left(1 + \frac{1}{a} \right) - 1 \quad (162)$$

$$0 < \sqrt{1 - \frac{4}{r}} = a < 1, \quad \spadesuit = \int_0^1 dx \ln(x-a) \quad (163)$$

$$\heartsuit(r) = \spadesuit + \ln(1+a) + a \ln \left(1 + \frac{1}{a} \right) - 1 \quad (164)$$

$$\spadesuit = \int_0^a dx \ln(x-a) + \int_a^1 dx \ln(x-a) \quad (165)$$

$$\spadesuit = \int_0^a dx \ln(-|x-a|) + \int_a^1 dx \ln(x-a) \quad (166)$$

$$\spadesuit = \ln(-1)a + \int_0^a dx \ln|x-a| + \int_a^1 dx \ln(x-a) \quad (167)$$

$$\ln(-1) = \pm i\pi \quad (168)$$

$$\spadesuit = \pm i\pi a + \int_0^a dx \ln(a-x) + \int_a^1 dx \ln(x-a) \quad (169)$$

$$\int_0^a dx \ln(a-x) = a(\ln a - 1) \quad (170)$$

$$\int_a^1 dx \ln(x-a) = (1-a)[\ln(1-a) - 1]$$

$$\spadesuit = \pm i\pi a - a \ln \left(\frac{1}{a} - 1 \right) + \ln(1-a) - 1 \quad (171)$$

$$\heartsuit(r) = \pm i\pi a + a \ln \left(\frac{1+a}{1-a} \right) + \ln(1-a^2) - 2 \quad (172)$$

$$f(r) = \sqrt{1 - \frac{4}{r}} \quad (173)$$

$$\heartsuit(r) = \pm i\pi f(r) + f(r) \ln \left[\frac{1+f(r)}{1-f(r)} \right] + \ln \left(\frac{4}{r} \right) - 2 \quad (174)$$

$$\spadesuit(r) = f(r) \ln \left[\frac{1+f(r)}{1-f(r)} \right] - 2 \pm i\pi f(r) \quad (175)$$

10. $\chi \rightarrow \chi$ MATRIX ELEMENT: EXPLICIT FORM

$$\mathcal{M} = \frac{g^2}{8\pi^2} \left[\frac{2}{\epsilon} - \gamma + \ln \left(4\pi \frac{\beta^2}{m^2} \right) - \spadesuit \left(\frac{s}{m^2} \right) \right] \quad (176)$$

Only $\spadesuit$ can be imaginary.

10.1. $0 < s/m^2 = r \leq 4$

$$\spadesuit(r) = 2\sqrt{\frac{4}{r} - 1} \operatorname{asin} \left(\frac{\sqrt{r}}{2} \right) - 2 \quad (177)$$

In this range $\mathcal{M}$ is real.

10.2. $4 \leq s/m^2 = r$

$$f(r) = \sqrt{1 - \frac{4}{r}} \quad (178)$$

$$\spadesuit(r) = \operatorname{Re} \spadesuit(r) + i \operatorname{Im} \spadesuit(r) \quad (179)$$

$$\operatorname{Re} \spadesuit(r) = f(r) \ln \left[\frac{1+f(r)}{1-f(r)} \right] - 2 \quad (180)$$

$$\operatorname{Im} \spadesuit(r) = \pm \pi f(r) \quad (181)$$

11. DECAY: $\chi \rightarrow \phi\phi$

The tree level amplitude has one diagram (Fig. 6).

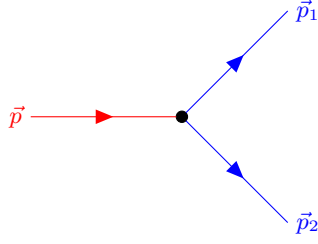
$$i\mathcal{M} = -2ig \implies \mathcal{M} = -2g \quad (182)$$

$$\vec{P} = \vec{p}_1 + \vec{p}_2, \quad \vec{p} = (E, \mathbf{p}), \quad \vec{p}_n = (E_n, \mathbf{p}_n) \quad (183)$$

$$|\mathcal{M}|^2 = 4g^2 \quad (184)$$

From Schwartz Eq. 5.24:

$$d\Gamma = \frac{|\mathcal{M}|^2}{2!2E} d\Pi_{\text{LIPS}} = \frac{|\mathcal{M}|^2}{4E} d\Pi_{\text{LIPS}}. \quad (185)$$


 FIG. 6. $i.\mathcal{M}$

I need 2! because the final state has 2 identical particles.
This equation only holds in the original particle's rest frame:

$$\vec{p} = (E, \mathbf{p}) = (E, \mathbf{0}). \quad (186)$$

From Schwartz Eq. 5.21:

$$d\Pi_{\text{LIPS}} = \frac{d^3 p_1}{2E_1(2\pi)^3} \frac{d^3 p_2}{2E_2(2\pi)^3} (2\pi)^4 \delta^{(4)}(\vec{p} - \vec{P}). \quad (187)$$

$$d\Pi_{\text{LIPS}} = \frac{d^3 p_1 d^3 p_2}{16\pi^2 E_1 E_2} \delta^{(4)}(\vec{p} - \vec{P}) \quad (188)$$

$$d\Gamma = \frac{|\mathcal{M}|^2 d^3 p_1 d^3 p_2}{64\pi^2 E E_1 E_2} \delta^{(4)}(\vec{p} - \vec{P}) \quad (189)$$

$$d\Gamma = \frac{4g^2 d^3 p_1 d^3 p_2}{64\pi^2 E E_1 E_2} \delta^{(4)}(\vec{p} - \vec{P}) = \frac{g^2 d^3 p_1 d^3 p_2}{16\pi^2 E E_1 E_2} \delta^{(4)}(\vec{p} - \vec{P}) \quad (190)$$

$$s = p^2 = (E, \mathbf{0})^2 = E^2 - \mathbf{0}^2 = E^2 \implies E = \sqrt{s} \quad (191)$$

$$d\Gamma = \frac{g^2 d^3 p_1 d^3 p_2}{16\pi^2 \sqrt{s} E_1 E_2} \delta^{(4)}(\vec{p} - \vec{P}) \quad (192)$$

$$16\pi^2 \sqrt{s} \Gamma = g^2 \int d^3 p_1 d^3 p_2 \frac{\delta^{(4)}(\vec{p} - \vec{P})}{E_1 E_2} \quad (193)$$

$$16\pi^2 \sqrt{s} \Gamma = g^2 \int d^3 p_1 d^3 p_2 \frac{\delta^{(4)}(\vec{p} - \vec{p}_1 - \vec{p}_2)}{E_1 E_2} \quad (194)$$

$$16\pi^2 \sqrt{s} \Gamma = g^2 \int d^3 p_1 d^3 p_2 \frac{\delta(E - E_1 - E_2)}{E_1 E_2} \delta^{(3)}(\mathbf{p} - \mathbf{p}_1 - \mathbf{p}_2) \quad (195)$$

$$\mathbf{p} = \mathbf{0}, \quad \delta(x) = \delta(-x) \quad (196)$$

$$16\pi^2 \sqrt{s} \Gamma = g^2 \int d^3 p_1 d^3 p_2 \frac{\delta(E - E_1 - E_2)}{E_1 E_2} \delta^{(3)}(\mathbf{p}_1 + \mathbf{p}_2) \quad (197)$$

$$\vec{p}_n = (E_n, \mathbf{p}_n) \implies p_n^2 = m^2 = E_n^2 - \mathbf{p}_n^2 \implies E_n = \sqrt{m^2 + \mathbf{p}_n^2} \quad (198)$$

$$16\pi^2 \sqrt{s} \Gamma = g^2 \int d^3 p_1 d^3 p_2 \frac{\delta(E - E_1 - E_2) \delta^{(3)}(\mathbf{p}_1 + \mathbf{p}_2)}{\sqrt{m^2 + \mathbf{p}_1^2} \sqrt{m^2 + \mathbf{p}_2^2}} \quad (199)$$

$$16\pi^2 \sqrt{s} \Gamma = g^2 \int d^3 p_1 d^3 p_2 \delta^{(3)}(\mathbf{p}_1 + \mathbf{p}_2) \frac{\delta\left(\sqrt{s} - \sqrt{m^2 + \mathbf{p}_1^2} - \sqrt{m^2 + \mathbf{p}_2^2}\right)}{\sqrt{m^2 + \mathbf{p}_1^2} \sqrt{m^2 + \mathbf{p}_2^2}} \quad (200)$$

$$16\pi^2 \sqrt{s} \Gamma = g^2 \int d^3 p_2 \frac{\delta\left(\sqrt{s} - \sqrt{m^2 + (-\mathbf{p}_2)^2} - \sqrt{m^2 + \mathbf{p}_2^2}\right)}{\sqrt{m^2 + (-\mathbf{p}_2)^2} \sqrt{m^2 + \mathbf{p}_2^2}} \quad (201)$$

$$16\pi^2 \sqrt{s} \Gamma = g^2 \int d^3 p_2 \frac{\delta\left(\sqrt{s} - \sqrt{m^2 + \mathbf{p}_2^2} - \sqrt{m^2 + \mathbf{p}_2^2}\right)}{\sqrt{m^2 + \mathbf{p}_2^2} \sqrt{m^2 + \mathbf{p}_2^2}} \quad (202)$$

$$16\pi^2 \sqrt{s} \Gamma = g^2 \int d^3 p_2 \frac{\delta\left(\sqrt{s} - 2\sqrt{m^2 + \mathbf{p}_2^2}\right)}{m^2 + \mathbf{p}_2^2} \quad (203)$$

$$16\pi^2 \sqrt{s} \Gamma = g^2 \int d^3 q \frac{\delta\left(\sqrt{s} - 2\sqrt{m^2 + \mathbf{q}^2}\right)}{m^2 + \mathbf{q}^2} \quad (204)$$

$$16\pi^2 \sqrt{s} \Gamma = g^2 \int_0^{4\pi} d\Omega \int_0^\infty dq q^2 \frac{\delta\left(\sqrt{s} - 2\sqrt{m^2 + q^2}\right)}{m^2 + q^2} \quad (205)$$

$$16\pi^2 \sqrt{s} \Gamma = 4\pi g^2 \int_0^\infty dq q^2 \frac{\delta\left(\sqrt{s} - 2\sqrt{m^2 + q^2}\right)}{m^2 + q^2} \quad (206)$$

$$4\pi \sqrt{s} \Gamma = g^2 \int_0^\infty dq q^2 \frac{\delta\left(\sqrt{s} - 2\sqrt{m^2 + q^2}\right)}{m^2 + q^2} \quad (207)$$

$$q = xm, \quad x = \frac{q}{m}, \quad x(0) = 0, \quad x(\infty) = \infty, \quad dq = m dx \quad (208)$$

$$4\pi \sqrt{s} \Gamma = g^2 m \int_0^\infty dx x^2 \frac{\delta\left(\sqrt{s} - 2m\sqrt{1 + x^2}\right)}{1 + x^2} \quad (209)$$

For an on-shell χ particle:

$$s = p^2 = \mu^2. \quad (210)$$

$$4\pi\mu\Gamma = g^2m \int_0^\infty dx x^2 \frac{\delta[h(x)]}{1+x^2} \quad (211)$$

$$h(x) = \mu - 2m\sqrt{1+x^2} \quad (212)$$

$$h(x_0) = 0 \quad \& \quad 0 \leq x_0 \implies x_0 = \sqrt{\left(\frac{\mu}{2m}\right)^2 - 1} \quad (213)$$

Note $\mu \geq 2m$, otherwise the decay is impossible.

$$\delta[h(x)] = \frac{\delta(x-x_0)}{|h'(x_0)|} \quad (214)$$

$$4\pi\mu\Gamma = \frac{g^2m}{|h'(x_0)|} \int_0^\infty dx \frac{x^2}{1+x^2} \delta(x-x_0) \quad (215)$$

$$4\pi\mu\Gamma = \frac{g^2m}{|h'(x_0)|} \frac{x_0^2}{1+x_0^2} \quad (216)$$

$$h'(x) = -\frac{2xm}{\sqrt{1+x^2}} \quad (217)$$

$$|h'(x_0)| = \frac{2x_0m}{\sqrt{1+x_0^2}}, \quad x_0 \geq 0 \quad (218)$$

$$\Gamma = \frac{g^2}{8\pi\mu} \frac{x_0}{\sqrt{1+x_0^2}} \quad (219)$$

$$\Gamma = \frac{g^2m}{4\pi\mu^2} \sqrt{\left(\frac{\mu}{2m}\right)^2 - 1} \quad (220)$$

$$\boxed{\Gamma = \frac{g^2}{8\pi\mu} \sqrt{1 - \left(\frac{2m}{\mu}\right)^2}} \quad (221)$$

12. OPTICAL THEOREM

Schwartz Eq. 24.13 gives

$$\text{Im } \mathcal{M} = \mu\Gamma. \quad (222)$$

$$\text{Im } \mathcal{M} = \frac{g^2}{8\pi} \sqrt{1 - \left(\frac{2m}{\mu}\right)^2} \quad (223)$$

$$\mathcal{M} = \frac{g^2}{8\pi^2} \left[\frac{2}{\epsilon} - \gamma + \ln\left(4\pi \frac{\beta^2}{m^2}\right) - \spadesuit\left(\frac{s}{m^2}\right) \right] \quad (224)$$

$$\text{Im } \mathcal{M} = \text{Im} \frac{g^2}{8\pi^2} \left[\cancel{\frac{2}{\epsilon}} - \gamma + \ln\left(\cancel{4\pi \frac{\beta^2}{m^2}}\right) - \spadesuit\left(\frac{s}{m^2}\right) \right] \quad (225)$$

$$\text{Im } \mathcal{M} = -\frac{g^2}{8\pi^2} \text{Im } \spadesuit\left(\frac{s}{m^2}\right) \quad (226)$$

$$\text{Im } \spadesuit(r) = \pm\pi f(r) \quad (227)$$

$$\text{Im } \mathcal{M} = \pm \frac{g^2}{8\pi^2} \pi f\left(\frac{s}{m^2}\right) = \pm \frac{g^2}{8\pi} f\left(\frac{s}{m^2}\right) \quad (228)$$

$$\text{Im } \mathcal{M} = \frac{g^2}{8\pi} \sqrt{1 - \left(\frac{2m}{\mu}\right)^2} \quad (229)$$

$$\pm \frac{g^2}{8\pi} f\left(\frac{s}{m^2}\right) = \frac{g^2}{8\pi} \sqrt{1 - \left(\frac{2m}{\mu}\right)^2} \quad (230)$$

$$\pm f\left(\frac{s}{m^2}\right) = \sqrt{1 - \left(\frac{2m}{\mu}\right)^2} \quad (231)$$

$$s = \mu^2, \quad f(r) = \sqrt{1 - \frac{4}{r}} \quad (232)$$

$$\pm f\left(\frac{\mu^2}{m^2}\right) = \sqrt{1 - \left(\frac{2m}{\mu}\right)^2} \quad (233)$$

$$x = \frac{\mu^2}{m^2} \quad (234)$$

$$\pm f(x) = \sqrt{1 - \frac{4}{x}} \quad (235)$$

$$\pm \cancel{f(x)} = \cancel{f(x)} \implies \pm 1 = 1 \quad (236)$$

I have confirmed the optical theorem up to an overall sign.
I explain the sign ambiguity above (see Eq. 89).